

Q1: Answer the following questions referencing the four given inverter circuits:

a) Which one of the circuits consumes static power when the input is high?

Circuit 1, 2, and 3 consume static power when input is high, as they all still allow current to flow from VDD to Vout to GND in this state.

b) Which one of the circuits consumes static power when the input is low?

None of them, they all cut the current between Vout and GND when the input is low

c) VOH of which circuit(s) is 1.2 V?

Circuits 1, 3 and 4. The top NMOS of 2 creates a drop of V_T , but all the others allow current to flow into Vout until it reaches the full VDD.

d) VOL of which circuit(s) is 0 V?

Circuit 4. Similar to question 1, all the others still allow current to flow from VDD to Vout. That current needs to be drained through the bottom NMOS, creating some non zero voltage across it.

e) The proper functionality of which circuit(s) depends on the size of the devices?

Circuits 2 and 3. They rely on one NMOS constantly powered and one NMOS switching, leaving Vout to be determined entirely by the size difference of the two. Circuits 1 and 4 are still affected by size, but should generally work okay regardless.

Q2: Calculate the V_{OH} and V_{OL} of the four inverters in the given figure by the means below:

a. By hand calculations using the 0.13 μm process parameters.

i. Use short-channel MOSFET equations where applicable.

$$k_n = \mu_n C_{ox} W/L$$

$$k_p = \mu_p C_{ox} W/L$$

if $V_{GS} < V_T$: subthreshold conduction region, $I_{DS}=0$

if $V_{GS} \geq V_T$:

if $V_{DS} < (V_{GS}-V_T)$: linear region, $I_{DS} = k/2 (2 (V_{GS}-V_T) V_{DS} - V_{DS}^2)$

if $V_{DS} \geq (V_{GS}-V_T)$: saturation region, $I_{DS} = k/2 (V_{GS}-V_T)^2 (1 + \lambda V_{DS})$

C1) $V_{OH} = 1.2\text{V}$, $V_{OL}=34.43\text{mV}$

$V_{GS}=V_{in}=0\text{V} < V_T=0.4$, subthreshold conduction region, $I_{DS}=0$, $V_{OH}=1.2\text{V}$

$V_{OH} = 1.2\text{V}$

$V_{GS}=V_{in}=1.2\text{V} > V_T=0.4$

$V_{DS}=V_{OL}\approx 0 < V_{GS}-V_T=0.8$ linear region

$$I_{DS} = k/2 (2 (V_{GS}-V_T) V_{DS} - V_{DS}^2)$$

$$k_n = 270 * 1.6 \times 10^{-6} * 1/0.1 = 4.32 \times 10^{-3}$$

$$V_{DS}=V_{OL}$$

$$I_{DS} = (V_{DD}-V_{OL})/R$$

$$(V_{DD}-V_{OL})/R = k/2 (2 (V_{GS}-V_T) V_{OL} - V_{OL}^2)$$

$$(1.2-V_{OL})/10\text{k} = (4.32 \times 10^{-3})/2 (2 (1.2-0.4) V_{OL} - V_{OL}^2)$$

$$(1.2-V_{OL}) = 21.6 (1.6 V_{OL} - V_{OL}^2)$$

$$1.2-V_{OL} = 34.56 V_{OL} - 21.6 V_{OL}^2$$

$$1.2 = 35.56 V_{OL} - 21.6 V_{OL}^2$$

$$V_{OL}=34.43\text{mV}$$

C2) $V_{OH}=0.8\text{V}$, $V_{OL}=31.57\text{mV}$

$V_{GS}=V_{in}=0\text{V} < V_T=0.4$, subthreshold conduction region, $I_{DS}(\text{driver})=0$,

Load NMOS turns off when $V_{GS}(\text{load})=V_{DD}-V_{OH}=V_T$

$$V_{OH}=V_{DD}-V_T=1.2-0.4=0.8\text{V}$$

$$V_{OH}=0.8\text{V}$$

$V_{GS}=V_{in}=1.2\text{V}$

$$V_{OL}\approx 0$$

$V_{DS}(\text{load})=V_{DD}-V_{OL}\approx 1.2 > V_{GS}-V_T=V_{DD}-V_T=0.8$, saturation region

$V_{DS}(\text{driver})=V_{OL}\approx 0 < V_{GS}-V_T=0.8$, linear region

equate currents

$$I_{DS}(\text{load}, \text{sat})=I_{DS}(\text{driver}, \text{lin})$$

$$k_{nL} = 0.432 \times 10^{-3}$$

$$k_{nD} = 5.184 \times 10^{-3}$$

$$k/2 (V_{GS}-V_T)^2 (1 + \lambda V_{DS}) = k/2 (2 (V_{GS}-V_T) V_{DS} - V_{DS}^2)$$

$$(0.432 \times 10^{-3})/2 ((1.2-VOL)-0.4)^2 (1 + 0.2 (1.2-VOL)) = (5.184 \times 10^{-3})/2 (2 (1.2-0.4) VOL - VOL^2)$$

$$0.432 (0.8-VOL)^2 (1.24 - 0.2 \cdot VOL) = 5.184 (1.6 VOL - VOL^2)$$

$$VOL = 38.16 \text{ mV}$$

C3) $VOH = 1.2 \text{ V}$, $VOL = 82.31 \text{ mV}$

$VGS = Vin = 0 \text{ V} < VT = 0.4$, subthreshold conduction region, $IDS(\text{driver}) = 0$,

Load NMOS turns off when $VGS(\text{load}) = VGG - VOH = VT$

$VOH = VGG - VT = 1.6 - 0.4 = 1.2 \text{ V}$ (and couldn't go any higher even with higher VGG)

$VOH = 1.2 \text{ V}$

$VGS = Vin = 1.2 \text{ V}$

$VOL \sim 0$

$VDS(\text{load}) = VDD - VOL \sim 1.2 = VGS - VT = VGG - VOL - VT \sim 1.2$, saturation region

NOTE: with $VT = 0.4 \text{ V}$, this only brings $VGS - VT$ up to VDS , the edge of the saturation and linear region. According to lab1 slides, if equal use saturation region, but I just did that in the last example, and this is literally called a *linear* enhancement-load inverter, so I'm going to just use the linear region equation for this

$VDS(\text{driver}) = VOL \sim 0 < VGS - VT = 0.4$, linear region

equate currents

$IDS(\text{load}, \text{lin}) = IDS(\text{driver}, \text{lin})$

$k_{nL} = 0.432 \times 10^{-3}$

$k_{nD} = 4.32 \times 10^{-3}$

$k/2 (2 (VGS - VT) VDS - VDS^2) = k/2 (2 (VGS - VT) VDS - VDS^2)$

$(2 (1.6 - VOL - 0.4) (1.2 - VOL) - (1.2 - VOL)^2) = 10 (2 (1.2 - 0.4) VOL - VOL^2)$

$(2 (1.2 - VOL) (1.2 - VOL) - (1.2 - VOL)^2) = 10 (1.6 VOL - VOL^2)$

$(1.2 - VOL)^2 = 16 VOL - 10 VOL^2$

$(1.2 - VOL)^2 = 16 VOL - 10 VOL^2$

$0 = 18.4 VOL - 11 VOL^2 - 1.44$

$VOL = 82.31 \text{ mV}$

C4) $VOH = 1.2 \text{ V}$, $VOL = 0 \text{ V}$

Both situations have one in the subthreshold conduction region, so the other is free to pull up/down all the way to VDD/gnd .

- b. By LTspice by performing a transient simulation on all four circuits.
- Use the updated SPICE level 49 NMOS & PMOS.txt file provided on LMS in the Labs folder.
 - Transient simulation duration is to be set to 10 μ s.
 - Place a 1 pF load capacitor out the output node of each inverter.
 - Read and follow the LTspice instructions in the following section to ensure correct simulation.

